

Parallel Binary Refinement of Labelled Set Partitions: Exact Kernels, Spectrum, and the Last-Collision Threshold

Route-A Dynamics Working Note C301

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Abstract

We solve a synchronous fragmentation chain on the labelled set partitions of $[n]$. At every step each label receives a fresh fair bit and each current block is refined by its two nonempty bit fibres. A coupling by binary words gives the full t -step kernel from every starting partition, the complete law of the number of blocks, and the exact absorption-time distribution. The transition matrix has eigenvalues 2^{k-n} with Stirling multiplicities $S(n, k)$.

1 The labelled parallel-fragmentation chain

Let $n \geq 1$ and let $\mathfrak{P}_n$ be the set of labelled set partitions of $[n] = \{1, \dots, n\}$. Write $\pi \preceq \sigma$ when σ refines π . Starting from $\Pi_0 = \pi$, every label independently receives a fresh bit in $\{0, 1\}$. In each block of π , labels with equal bits stay together; empty fibres are deleted. The resulting partition is Π_1 . Updates use fresh independent bits.

This is a synchronous chain: all blocks are marked in parallel. Labels are part of the state. In particular, quotienting by block sizes produces a different chain and different spectral multiplicities. Put

$$(q)_r = q(q-1) \cdots (q-r+1), \quad (q)_0 = 1,$$

and let $S(n, k)$ denote a Stirling number of the second kind. For $\pi \preceq \sigma$, let $r_B(\pi, \sigma)$ be the number of σ -blocks contained in $B \in \pi$.

2 Complete law

Theorem 1 (all-parameter solution). *For every $n \geq 1$, $t \geq 0$, and $\pi, \sigma \in \mathfrak{P}_n$, with $q = 2^t$,*

$$K_n^t(\pi, \sigma) = \mathbf{1}_{\{\pi \preceq \sigma\}} \prod_{B \in \pi} \frac{(q)_{r_B(\pi, \sigma)}}{q^{|B|}}. \quad (1)$$

Consequently, the one-step probability equals $2^{|\pi|-n}$ precisely when σ refines π and each π -block contains at most two σ -blocks, and equals zero otherwise.

From the one-block initial partition, for every $\sigma \in \mathfrak{P}_n$ and $1 \leq k \leq n$,

$$\Pr\{\Pi_t = \sigma\} = \frac{(2^t)_{|\sigma|}}{2^{tn}}, \quad (2)$$

$$\Pr\{|\Pi_t| = k\} = S(n, k) \frac{(2^t)_k}{2^{tn}}, \quad (3)$$

$$\mathbb{E}|\Pi_t| = 2^t [1 - (1 - 2^{-t})^n]. \quad (4)$$

Let $T_n = \min\{t : \Pi_t \text{ is discrete}\}$. Then

$$\Pr\{T_n \leq t\} = \frac{(2^t)_n}{2^{tn}}, \quad (5)$$

$$\Pr\{T_n = t\} = \frac{(2^t)_n}{2^{tn}} - \mathbf{1}_{\{t \geq 1\}} \frac{(2^{t-1})_n}{2^{(t-1)n}}, \quad (6)$$

$$\mathbb{E}T_n = \sum_{t \geq 0} \left[1 - \frac{(2^t)_n}{2^{tn}} \right]. \quad (7)$$

The series converges. The characteristic polynomial, source spectral determinant, and power traces are

$$\chi_{K_n}(x) = \prod_{k=1}^n (x - 2^{k-n})^{S(n,k)}, \quad (8)$$

$$\det(I - zK_n) = \prod_{k=1}^n (1 - z2^{k-n})^{S(n,k)}, \quad (9)$$

$$\mathrm{tr}(K_n^t) = \sum_{k=1}^n S(n,k) 2^{t(k-n)}. \quad (10)$$

Moreover,

$$\prod_{k=1}^n (K_n - 2^{k-n}I) = 0, \quad (11)$$

so K_n is diagonalizable over $\mathbb{Q}$.

Finally, if $n_j \rightarrow \infty$, t_j are integers, and $n_j^2/2^{t_j} \rightarrow \lambda \in (0, \infty)$, then

$$\Pr\{T_{n_j} \leq t_j\} \longrightarrow e^{-\lambda/2}. \quad (12)$$

Thus T_n has a tight $O(1)$ window around $2 \log_2 n$.

3 Binary-word proof

After t updates, attach to label i the word formed by its successive bits. These words are independent and uniform in a set of size $q = 2^t$. Inside each starting block B , two labels are together at time t exactly when their words agree. To realize r_B prescribed terminal fibres one must injectively assign words to those fibres. There are $(q)_{r_B}$ assignments. Starting blocks use disjoint label sets, hence their counts multiply, proving (1). Taking $t = 1$ gives the stated one-step kernel.

For a one-block start, a fixed k -block partition has probability $(q)_k/q^n$. There are $S(n,k)$ such labelled partitions, yielding (2) and (3); the identity $\sum_k S(n,k)(q)_k = q^n$ also verifies normalization. Alternatively, $|\Pi_t|$ is the number of occupied boxes after n independent throws into q boxes. Summing the q occupancy indicators proves (4).

Absorption means that all n words are distinct, which proves (5); differencing successive distribution functions gives (6). The tail-sum formula gives (7). A union bound gives

$$\Pr\{T_n > t\} \leq \binom{n}{2} 2^{-t},$$

so the tail series converges. For $n = 1$, all formulas reduce to $T_1 = 0$. The contract begins at $n = 1$; the empty-set convention $n = 0$ is not used.

References

- [1] P. Diaconis, C. Y. A. Pang, and A. Ram, ‘‘Hopf algebras and Markov chains: two examples and a theory,’’ *Journal of Algebraic Combinatorics* (2014), doi:10.1007/s10801-013-0456-7; arXiv:1206.3620.